

Generative Modeling

Diffusion Models

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Laboratory of methods for big data analysis



LAMBDA • HSE

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In this Lecture

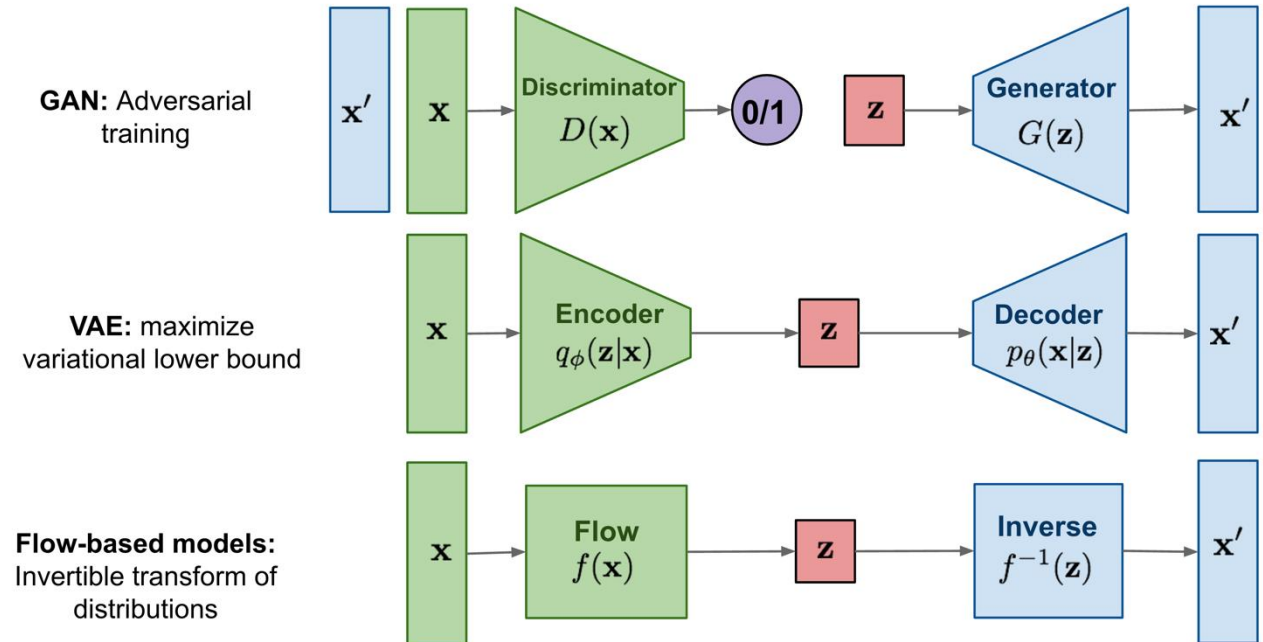
- ▶ Look back at the Generative Models
- ▶ Basic idea of Diffusion
- ▶ Training Diffusion Models
- ▶ Applications

Motivation

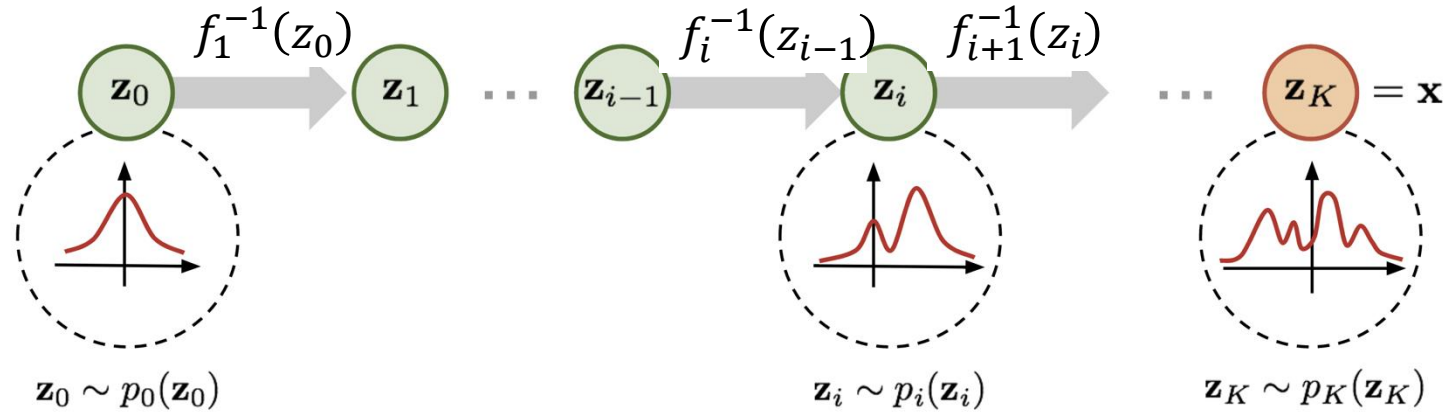


Look back

- ▶ Unstable training process of GANs
- ▶ Blurred and not realistic results of VAEs
- ▶ GANs and VAEs are overcomplicated (at least one extra network is used during the training process)
- ▶ Normalizing flows are limited due to fast Jacobian computation and invertibility



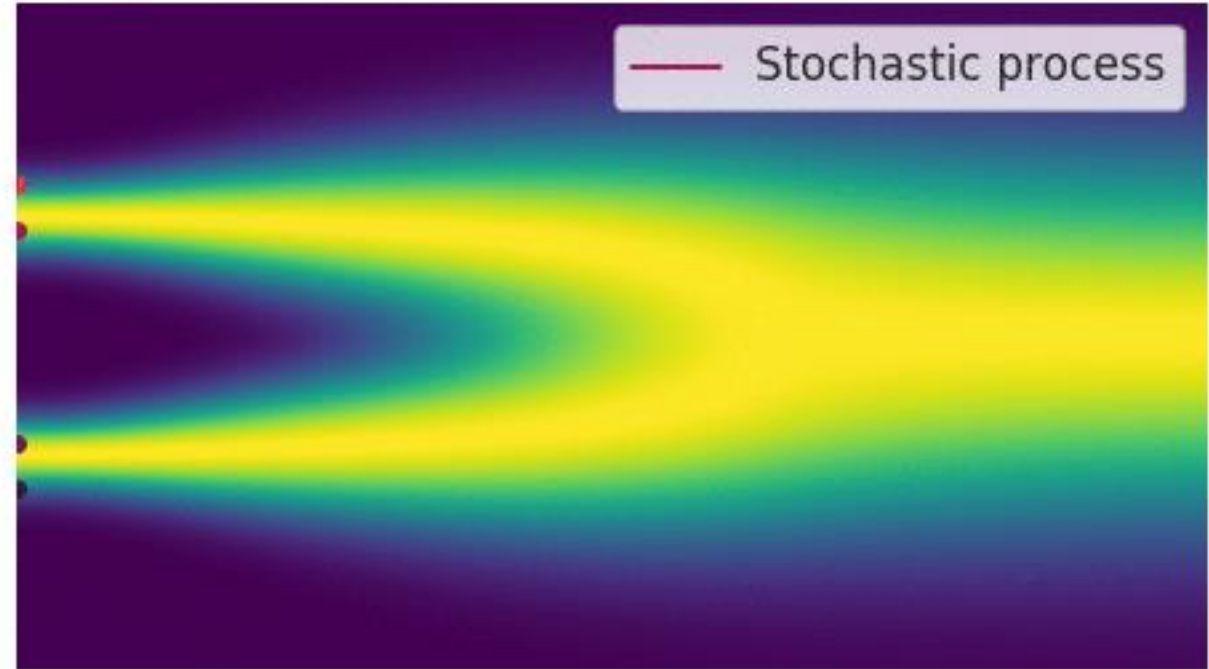
Flow discussion



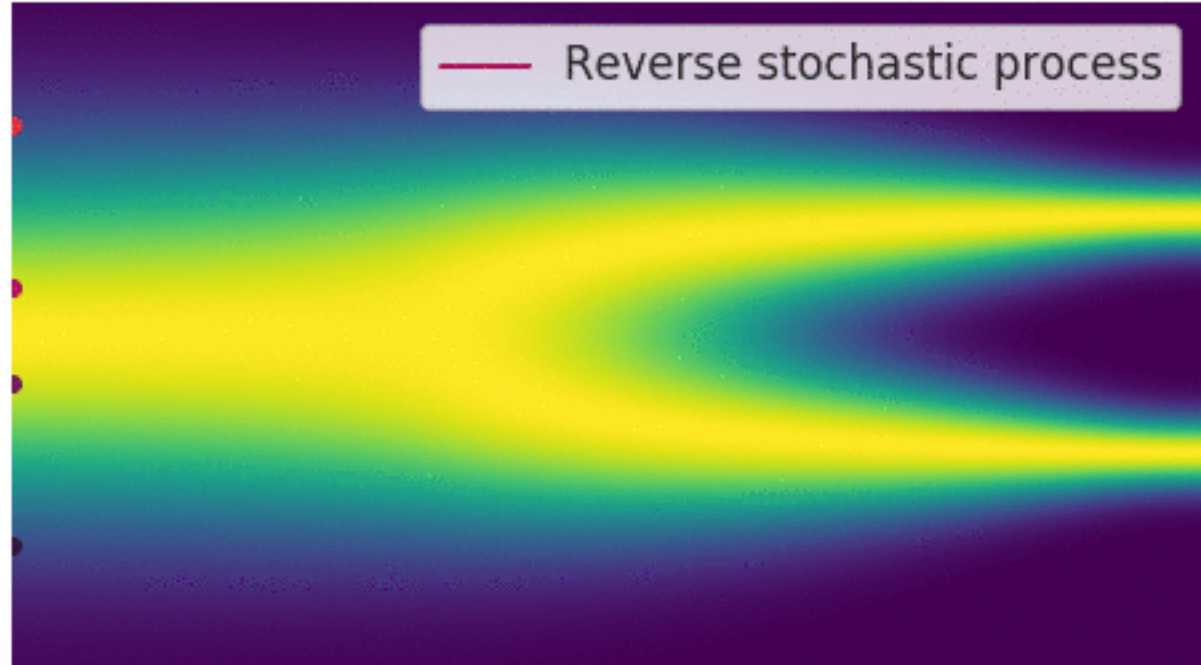
$$p_x(x) = p_{z_0}(f_K(\dots(f_1(x)))) \left| \det \frac{\partial f_1(z_1)}{\partial z_1} \right| \dots \left| \det \frac{\partial f_K(z_k)}{\partial z_k} \right|.$$

- ▶ It is possible to obtain $p_x(x)$ consecutively changing observables.
- ▶ The overall transformation is **invertible** if individual layers are invertible.
- ▶ Dimensions of each observable is the same.
- ▶ Fit is performed using ML estimate.
- ▶ **Need to calculate determinant.**

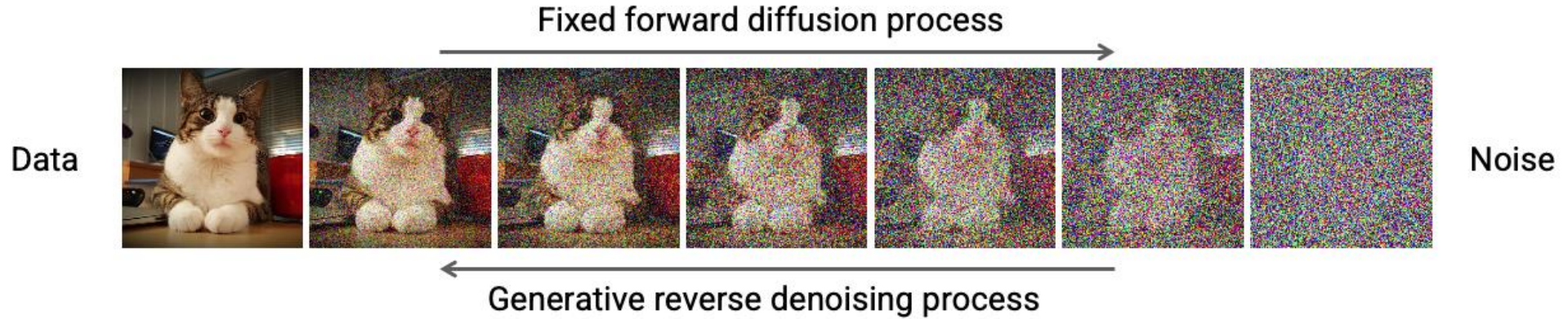
Forward diffusion



Reverse diffusion



Diffusion process

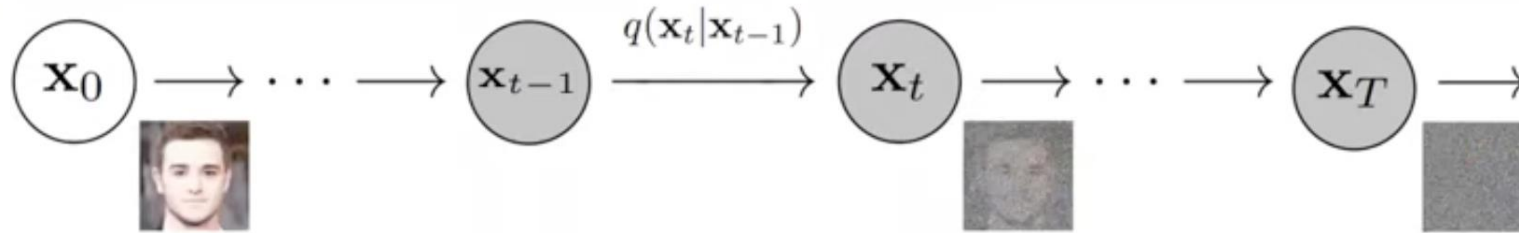


- ▶ Forget invertibility
- ▶ Fix forward process
- ▶ Learn approximate likelihood estimation

Diffusion Models



Forward Diffusion



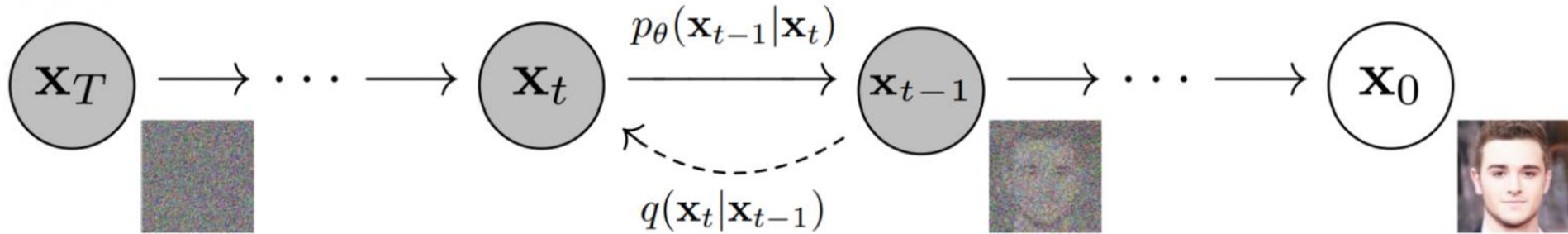
- ▶ $x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$
- ▶ $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$
- ▶ $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\epsilon\right) + \tilde{\beta}_tz$

$$\epsilon_t \sim N(0, I), \quad \alpha_t = 1 - \beta_t$$

$$\epsilon \sim N(0, I), \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

$$z \sim N(0, I), \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

Inverse Diffusion

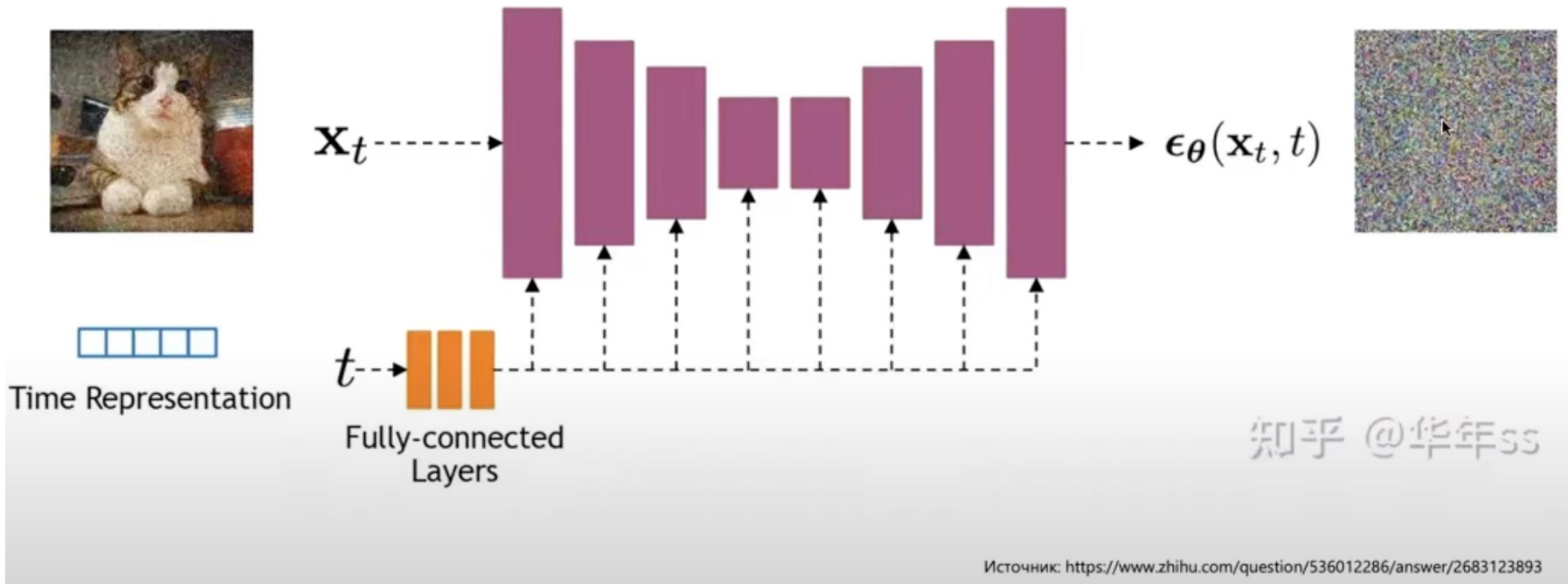


$$\hat{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z \quad z \sim N(0, I), \quad \sigma_t = \text{const}$$

Loss function:

$$L_t = \|x_{t-1} - \hat{x}_{t-1}\|_2^2 \propto \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \rightarrow \min_{\theta}$$

U-net architecture for diffusion



Training/Sampling

Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
       $\nabla_{\theta} \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}, t)\|^2$   
6: until converged
```

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```

<https://arxiv.org/abs/2006.11239>

Problem!

Problem: How to train p_θ ? Absolutely the same as with VAE!

$$\begin{aligned} -\log p_\theta(\mathbf{x}_0) &\leq -\log p_\theta(\mathbf{x}_0) + D_{\text{KL}}(q(\mathbf{x}_{1:T}|\mathbf{x}_0) \| p_\theta(\mathbf{x}_{1:T}|\mathbf{x}_0)) \\ &= -\log p_\theta(\mathbf{x}_0) + \mathbb{E}_{\mathbf{x}_{1:T} \sim q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})/p_\theta(\mathbf{x}_0)} \right] \\ &= -\log p_\theta(\mathbf{x}_0) + \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} + \log p_\theta(\mathbf{x}_0) \right] \\ &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \end{aligned}$$

$$\text{Let } L_{\text{VLB}} = \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \geq -\mathbb{E}_{q(\mathbf{x}_0)} \log p_\theta(\mathbf{x}_0)$$

See Seminar for detail

Results

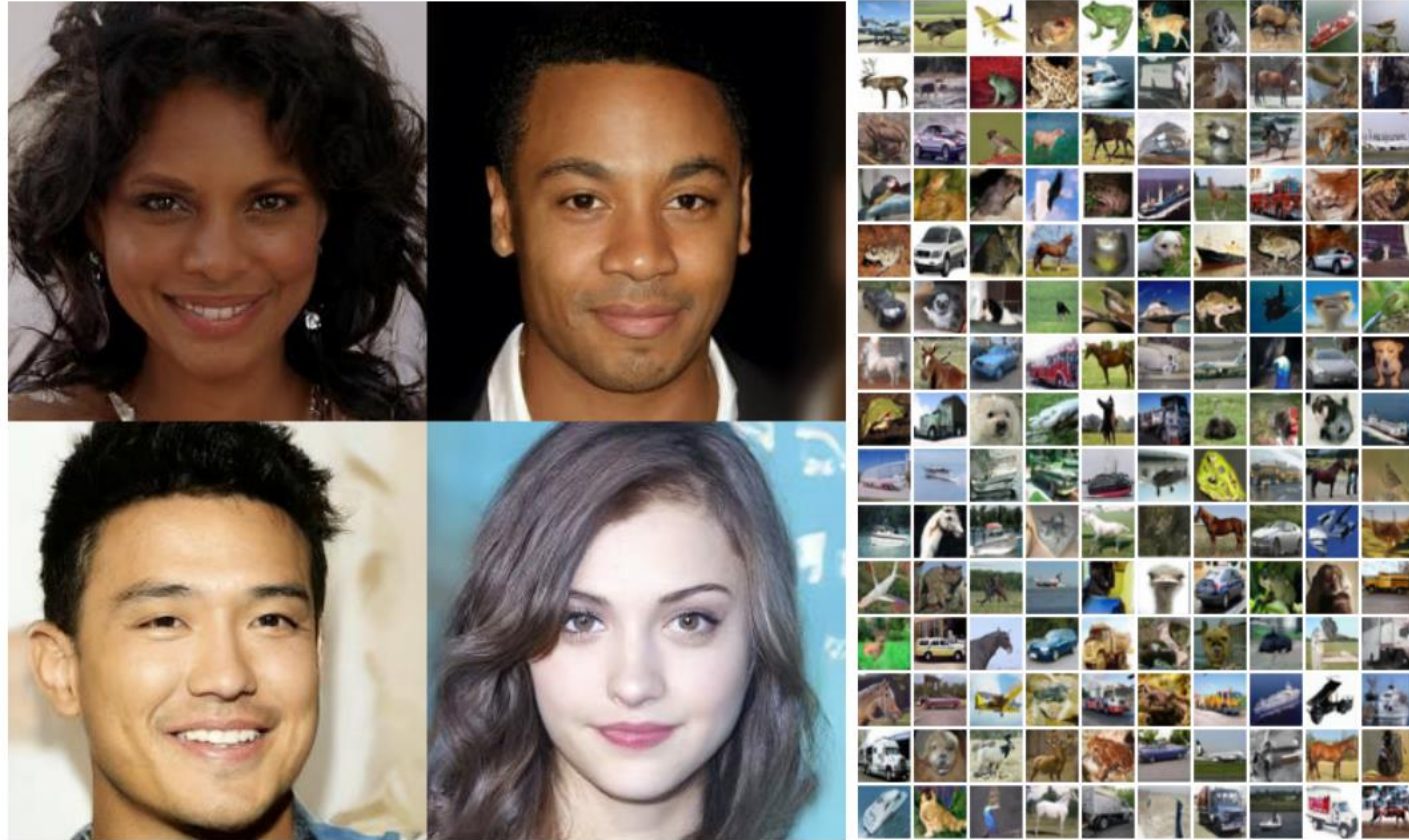


Figure 1: Generated samples on CelebA-HQ 256×256 (left) and unconditional CIFAR10 (right)

<https://arxiv.org/abs/2006.11239>

Conclusions

- ▶ Very Good Quality
- ▶ Closely related to Stochastic Differential Equations
- ▶ No Inner representation
- ▶ Should keep the same dimensions at each step

Conclusions



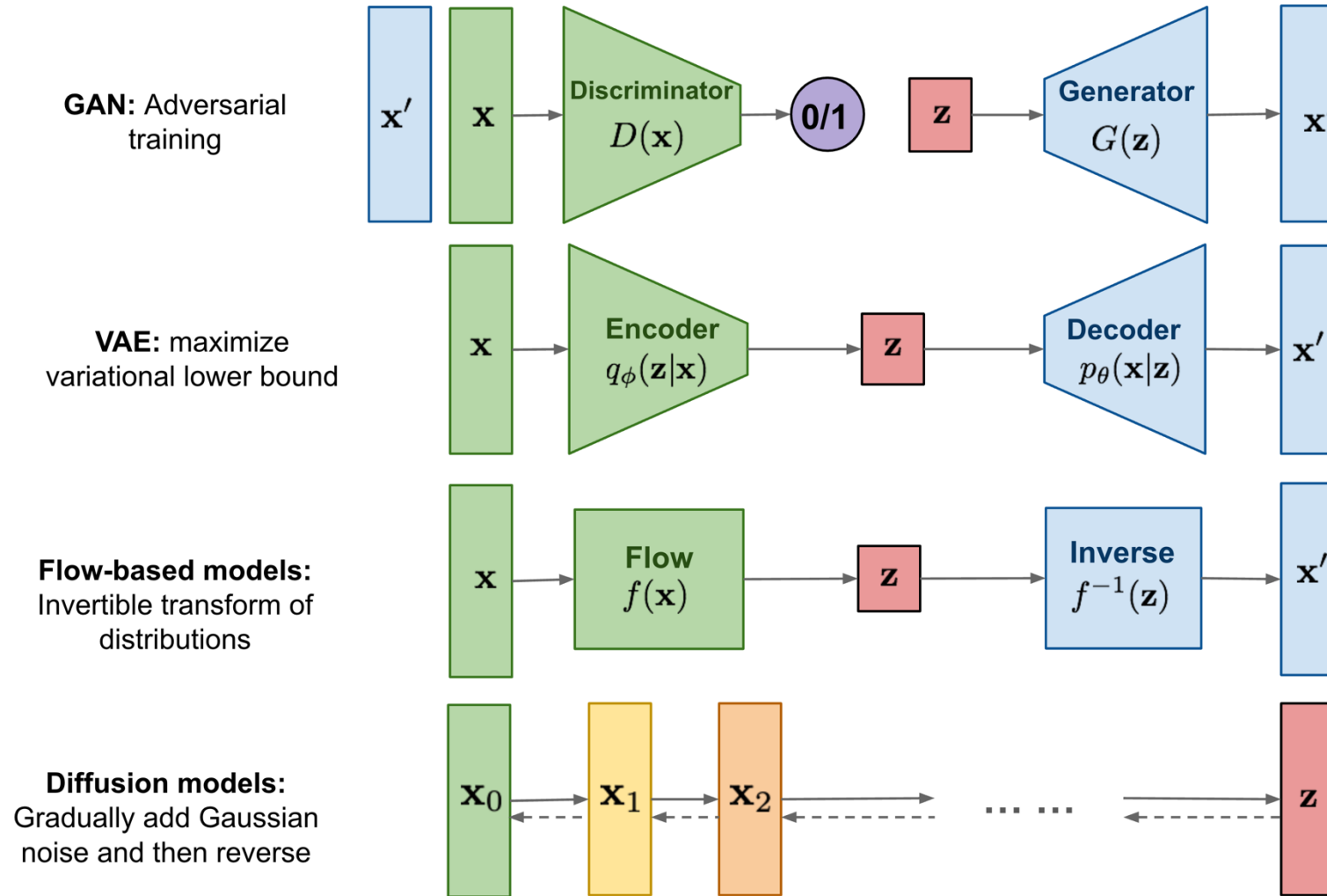
Generative model: Problem Statement

Three major tasks, given a generative model f from a class of models $\mathcal{F}$:

- ▶ **Estimation**: find the f in $\mathcal{F}$ that best matches observed data.
- ▶ **Evaluate Likelihood**: compute $f(z)$ for a given z .
- ▶ **Sampling**: drawing from f .

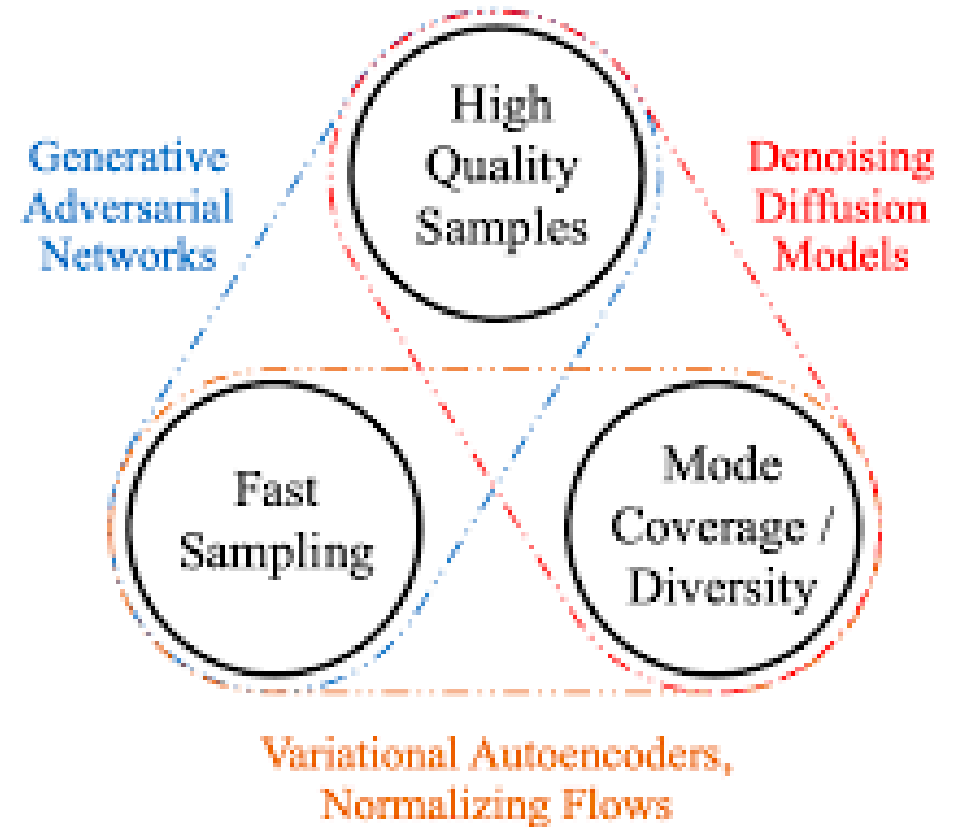
S. Nowozin et al. f-GAN: Training Generative Neural Samplers using Variational Divergence Minimization

Generative Modeling



Generative Trilemma

- **Estimation**: find the f in $\mathcal{F}$ that best matches observed data.
- **Evaluate Likelihood**: compute $f(z)$ for a given z .
- **Sampling**: drawing from f .



<https://arxiv.org/abs/2112.07804>